

Review

Catalytic Biomass Gasification for Syngas Production: Recent Progress in Tar Reduction and Future Perspectives

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Abstract

Biomass gasification is an effective process for converting organic wastes into syngas. Syngas is a biofuel that possesses several potential applications in the energy sector. However, the major bottleneck for the commercialization of this technology is tar production in biomass gasification, which affects gasifier performance and syngas yield/quality. Tar can be destructed by adopting in situ or ex situ modes of utilizing catalysts in biomass gasification. The added advantage of tar reduction is enhanced syngas energy content. Despite their advantages, catalysts face challenges such as high costs, declining performance over time, and difficulties in regeneration and recycling. Deactivation can also occur due to poisoning, fouling, and carbon buildup. While some natural materials have been tested as alternative materials, the financial sustainability and affordability of catalysts remain crucial for large-scale syngas production. This paper offers an overview of tar reduction strategies and the role of various catalysts in the gasification process and future perspectives on catalyst development for biomass gasification.

Keywords: biomass; catalyst; gasifier; syngas; tar



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1. Introduction

Energy is fundamental to the growth of the economy, society, and human life. However, rising energy consumption is being driven by population growth and rapid economic development. The over-exploitation of conventional fossil fuels has triggered an energy crisis, extensive waste generation, resource depletion, and significant environmental challenges [1]. Biofuels derived from different biomass resources (food waste, animal waste, crop waste, etc.) offer a wide range of usage as fuels similar to petroleum fuels [2]. Further, they can reduce or replace conventional petroleum fuels to fulfil Sustainable Development Goal 7: affordable and clean energy. In addition, they may also lessen expenses, address environmental problems, and mitigate the energy crisis.

Annual agricultural waste production in India is approximately 350–990 million t, contributing to the global production of 4 billion t [3]. Biomass availability and renewable energy production potential vary between countries due to factors like topography, seasonality, biodiversity, resource accessibility, technology, and finances. It is estimated that by 2050, biomass will provide 3000 TWh of power and reduce annual CO₂ equivalent emissions by 1.3 billion t [4]. Several technologies are used to obtain energy from this kind

of waste. Biochemical and thermochemical processes are the primary biomass conversion processes available to transform these residues into valuable products. Generally, agro-residues are categorized as lignocellulosic biomass, which is predominantly subjected to thermochemical conversion for energy recovery [5].

The gasification process is one of the thermochemical conversion processes used to generate a gaseous biofuel called producer gas, synthesis gas, or syngas from biomass feedstock. Syngas is used directly as a fuel or synthesized for biochemical and fuel production [6]. The syngas composition contains 30 to 60% CO, 5 to 15% CO₂, 25 to 30% H₂, and 0 to 5% CH₄, with traces of water vapor and sulfur compounds. However, it contains contaminants such as tar, hydrogen sulfide, particulate matter, ammonia, and alkali metals. These contaminants may lead to clogging and corrosion problems in gas engines. At the same time, internal combustion engines require clean and tar-free syngas. Tar formation is an inevitable step in biomass gasification if feedstocks with higher volatile matter contents are used. The tar is corrosive in nature and causes operational issues for extended periods of gasifier operation [7]. Hence, tar removal is an essential step in the biomass gasification process. Catalysts are used in thermo/biochemical reactions to improve overall conversion efficiency and final product quality.

The aim of this review was to provide an overview of recent progress in catalytic biomass gasification, mainly focusing on tar reduction, syngas quality, and syngas production. The keywords “catalytic biomass gasification”, “syngas yield”, “tar cracking mechanisms”, and “biomass gasifier catalysts” from scientific databases such as Scopus and Google Scholar were used. The inclusion areas were biomass resources relevant to gasification, types of biomass gasifiers, tar cracking mechanisms, catalyst performance and types, catalysts’ influence on syngas yield and composition, and current status and future perspectives of catalytic biomass gasification.

2. Biomass Resources

Biomass is one of the earliest energy sources used by humans. Through photosynthesis, plants convert solar energy into chemical energy in the form of glucose or sugar, which is termed biomass energy [8]. Biomass is commonly used to generate biofuels such as bioethanol, biodiesel, charcoal, bio-oil, biogas, and syngas. Biomass is considered as a renewable resource that is carbon-neutral when compared to fossil fuels [9]. Biomass resources categorized based on their origin are illustrated in Figure 1.

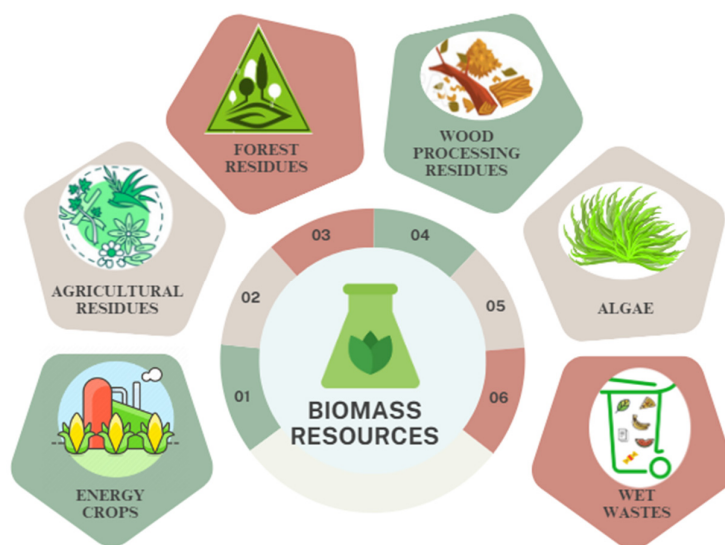


Figure 1. Types of biomass resources.

2.1. Energy Crops

Non-food, single-purpose energy crops can be planted on marginal land to produce biomass materials. Generally, herbaceous and woody are the two broad categories of energy crops. Perennial grasses are herbaceous energy crops, harvested once a year after reaching maximum productivity in two to three years. Examples are millet, miscanthus, bamboo, and sweet sorghum. Woody energy crops are fast-growing hardwood trees that can be harvested within 5 to 8 years after planting. Hybrid poplar, *Melia dubia*, and eucalypts are a few woody crops. These can enhance soil health, water quality, wildlife habitats, and agricultural productivity and diversify revenue streams [10,11]. They offer a high biomass yield and constant quality, making them appropriate for the gasification process; their high moisture content, ash content, and land use are some significant challenges for utilizing them in gasification.

2.2. Agricultural Residues

Agricultural residues such as straw, husks, and leaves are abundant, diverse, and widely scattered across cultivated areas. Paddy straw, barley straw, wheat straw, oat straw, sorghum stubble, and corn stalks are a few examples. Agricultural residues can be used for various applications. In addition, they are used directly as a source of nutrients for the next crop. Agricultural residues can be utilized as raw materials to produce biofuels/bioproducts/biochemicals, and they offer farmers a chance to earn additional income [12]. They are a cost-effective feedstock material for the gasification process with a sustainable disposal method; however, their composition inconsistency and high ash content cause slagging and fouling in operation.

2.3. Forest Residues

Forest residues fall under two categories, namely primary forest residues and secondary forest residues. Primary residues refer to forest leftovers from logging operations, while secondary residues come from biomass obtained from whole trees harvested for biomass or from the forest debris remaining after logging. Forest residues refer to the unhealthy, dead, deformed, or otherwise unmarketable parts of trees often left in forests. Reducing the risk of fire and pests while promoting resilience, productivity, and vitality in forests can be achieved by harvesting excess woody biomass and leftovers [13]. Forest residues have a good potential for use as a feedstock in gasification due to their low moisture content and low ash content, but they face challenges in collection, transport, and availability.

2.4. Wood Processing Residues

Wood processing residues refer to the waste streams and byproducts that are produced in the wood processing industry. Sawdust, bark, branches, and leaves/needles are byproducts of the processing of wood into products or pulp [13]. They are dry materials and an efficient feedstock for gasification. In contrast, limited supply and competition with other industrial uses lessen their availability for gasification.

2.5. Algae

Various highly productive organisms, such as blue-green algae, macroalgae (seaweed), and microalgae, are collectively known as algae and are used as a feedstock for bioenergy. Algae require sunlight and nutrients to produce algal biomass containing essential elements like lipids, carbohydrates, and proteins, which can be processed to produce different biofuels and bioproducts. Depending on the strain, algae can thrive in fresh/brackish/saline water from different aquatic sources [14]. After lipid extraction from algae, the residue

can be used as a feedstock for syngas production. Lipid-free algal cake can be used as a feedstock for syngas production [15]. It can be cultivated on non-arable land and produces a higher yield for utilization in gasification; still, its high moisture content and complexity in feedstock processing restrict its utilization.

2.6. Wet Waste

Food waste from industries, institutions, and homes; organic-rich solid wastes; animal dung; manure from concentrated livestock operations; and organic waste from industrial activities are examples of wet waste materials. Converting these wastes into electricity also supports rural economies by generating additional revenue with waste disposal [16]. Wet waste is abundant and can be utilized as gasification feedstock after dehydration, yet its high moisture content increases processing costs and complexity.

Embracing a diverse array of biomass resources supports sustainable energy practices, enhances environmental stewardship, and fosters economic opportunities globally.

2.7. Influence of Types of Biomass on Gasification Performance and Tar Formation

The type of biomass has a significant impact on the formation of tar during gasification, which affects the quality of syngas and the overall performance of the process. The components of biomass, viz., cellulose, hemicellulose, and lignin, degrade at different temperatures, resulting in diverse tar profiles. Lignin has a complex aromatic structure, and compared to the other components of biomass, it produces more tar molecules, especially polycyclic aromatic hydrocarbons (PAHs), which are more persistent and difficult to remove. On the other hand, at higher temperatures, cellulose and hemicellulose form lighter, less stable tar molecules that are more prone to breaking into syngas [17]. Furthermore, biomass moisture concentration is important because increased moisture content might lower gasification temperatures, promoting secondary processes that break down tar precursors and reduce tar production [18]. Thus, selecting biomass with a suitable moisture content and composition is crucial for improving gasification efficiency and tar reduction.

3. Biomass Gasifiers

The biomass gasification process produces syngas from biomass through a controlled supply of an oxidizing agent in a gasifier. Globally, an extensive range of gasifiers have been designed based on various feed materials, sizes, and syngas characteristics. Fixed bed, fluidized bed, and entrainment flow gasifiers are the three major types of gasifiers. Plasma and hydrothermal gasifiers are advanced gasifiers with high feedstock conversion efficiencies but are still under development and not commercially available. The primary variations of gasification technology are based on biomass input and gas flow, operating pressure, temperature, and oxidant selection (common oxidants: steam, air, and O₂; advanced oxidants: CO₂, H₂, and mixed oxidants). Based on process temperature, gasifiers are classified as low-temperature (<1000 °C) and high-temperature (>1200 °C) gasifiers. During the low-temperature gasification process, methane, higher-hydrocarbon tars, the valuable byproduct biochar, and ash along with syngas with a higher ratio of H₂ and CO₂ are produced. In contrast, high-temperature gasification minimizes the production of char, tar, and methane and includes a gas cleaning and regeneration system [19]. A schematic sketch of all types of gasifiers is presented in Figure 2.

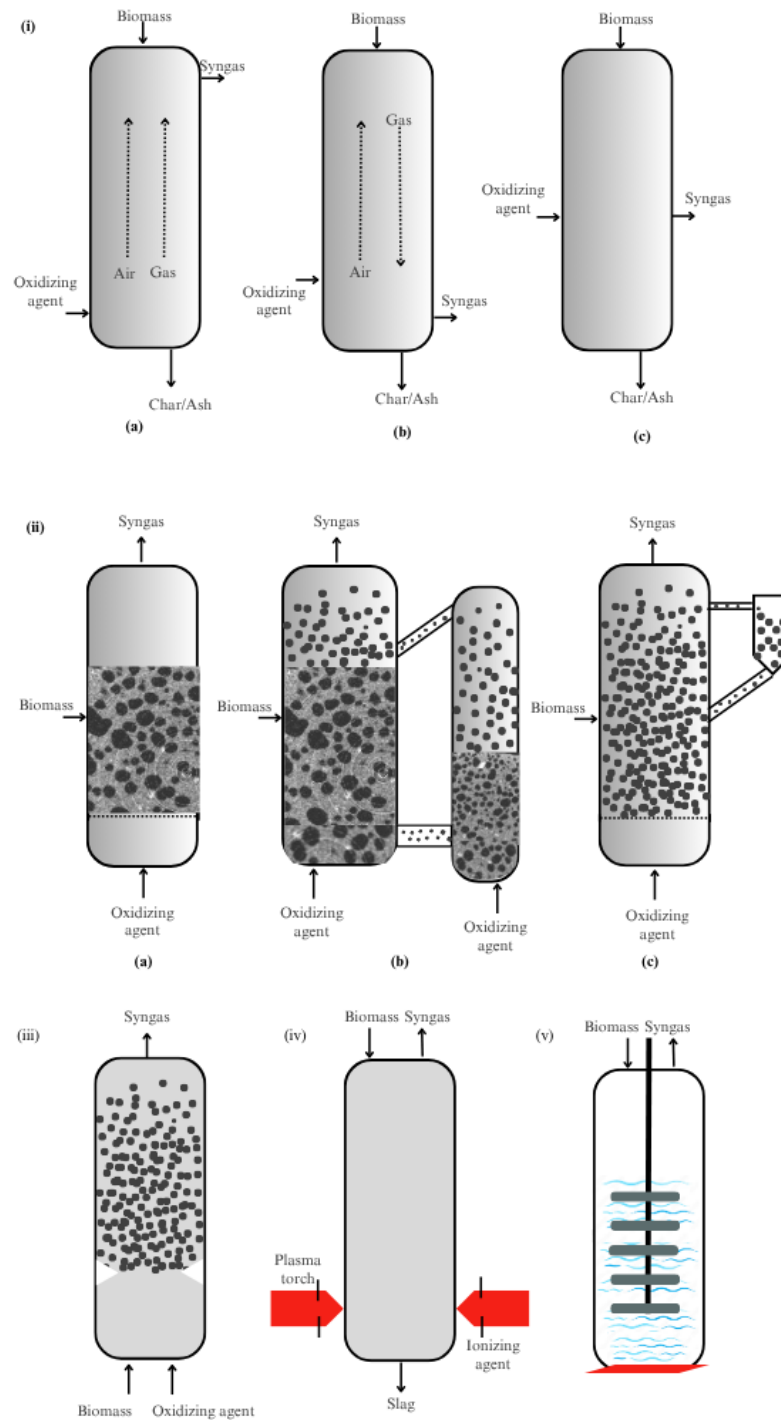


Figure 2. Different types of biomass gasifiers. (i) Fixed bed gasifier. (a) Updraft. (b) Downdraft. (c) Cross draft. (ii) Fluidized bed gasifier. (a) Bubbling. (b) Circulating. (c) Dual. (iii) Entrained flow gasifier. (iv) Plasma gasifier. (v) Hydrothermal gasifier.

3.1. Fixed Bed Gasifiers

The fuel bed in a fixed bed gasifier requires high permeability since gas moves at a slow rate. The oxidizing agent used can be either oxygen or air. A fixed bed gasifier has four distinct heat zones: pyrolysis, reduction, oxidation, and drying. In the drying zone, the residual moisture in the biomass evaporates. The biomass is heated to 400 °C in the pyrolysis zone without the addition of O₂, resulting in the production of gas, liquid hydrocarbon tar, and char. Char is converted to gaseous form in the gasification (reduction)

zone at more than 800 °C. The residual biomass char is oxidized in the oxidation zone at about 1000 °C and supplies the required thermal energy for the reactions in other zones [20].

The three main types of fixed bed gasifiers are downdraft, updraft, and crossdraft (Figure 2(i)). In updraft and downdraft gasifiers, fuel is fed into the top of the gasifier. In downdraft gasifiers, syngas exits from the bottom with the oxidizer introduced from above or the side, moving both syngas and biomass concurrently. In contrast, gas and biomass move in opposite directions in updraft gasifiers, with the oxidizer introduced from the bottom and syngas exiting from the top. As biomass descends, some of the char burns, providing more heat.

In a downdraft gasifier, tar can decompose at higher temperatures as it passes through the combustion zone [21]. Conversely, in an updraft gasifier, the gas exits through the pyrolysis zone, carrying the tar matter with the gas [22]. The downdraft gasifier can produce syngas with low tar content compared with different types of biomass gasifiers. It is well-suited due to its lower raw syngas temperatures and higher thermal efficiency. In a crossdraft gasifier, air is introduced from the side, while biomass is fed from the top of the gasifier. As the biomass passes through the gasifier, it undergoes drying, descends, and further undergoes gasification. Simultaneously, pyrolysis, oxidation, and reduction occur at the side where air is introduced into the gasifier [23,24]. The throat system in the gasifier is vital for destroying tar generated during gasification. The throat system typically restricts the largest gasifier size to 10 MW_{th}. However, a set of gasifier systems can be used to achieve higher capacities. When there is partial load operation or unstable operation, excessive tar production occurs [25]. When biochar is considered as a desired product in fixed bed gasifiers, updraft gasifiers usually yield high biochar. In contrast, downdraft gasifiers and crossdraft gasifiers yield less biochar due to high operating temperatures and complete biomass conversion to gas.

3.2. Fluidized Bed Gasifiers

In fluidized bed gasifiers, finely powdered biomass particles are suspended with a large quantity of inert bed material (silica sand/fly ash/dolomite) with higher gas flow. The gas flow rate ensures that particles undergoing gasification mix with fresh feedstock particles. Ash may either agglomerate or be discharged in dry form. These gasifiers are operated at comparatively lower temperatures (<900 °C), allowing for reactive feedstock utilization. Few gasifiers are intended to operate under pressure. Although they typically use air for fluidization due to the large gas volumes required, oxygen-blown systems are also feasible [26]. Fluidized bed gasifiers are smaller than fixed bed gasifiers due to their higher reaction rates and efficient heat exchange facilitated by intense bed mixing [27]. However, varying residence times of particles can lead to incomplete carbon burnout, which is a limitation of fluidized bed gasifiers [28]. These gasifiers generally yield less biochar compared to the fixed bed type due to more complete gasification and high reaction uniformity. The various types of fluidized bed gasifiers (Figure 2(ii)) are as follows.

3.2.1. Bubbling Fluidized Bed (BFB) Gasifiers

In a BFB gasifier, air, O₂, or steam is injected upward through the bed at a velocity sufficient to maintain material movement (typically 1 to 3 m.s⁻¹) while biomass is introduced from the side. When oxygen is used as the oxidizing agent instead of air, high-quality syngas is produced. A cyclone positioned near the syngas outlet collects unburnt char and ash particles [29].

3.2.2. Circulating Fluidized Bed (CFB) Gasifiers

In a CFB gasifier, the biomass particles are in a suspension with higher air, O₂, and steam velocity (usually 5–10 ms⁻¹). Particles are recirculated back into the fluidized bed

using a siphon and a cyclone. Higher velocities lead to higher particle concentrations and attrition. Consequently, gasifiers must be designed to withstand erosion caused by fast-moving particles. CFB gasifiers typically handle various feedstocks when the feedstock size is maintained at a size of less than 20 mm. The cyclone separates ash and bed material and recirculates to the reactor [30].

3.2.3. Dual Fluidized Bed (DFB) Gasifiers

A DFB gasifier consists of two important types of fluidized beds used in the process (bubbling fluidized bed gasifier and circulating fluidized bed combustor). The fluidized bed gasifier involves generating the syngas and char. In contrast, the fluidized bed combustor deals with the oxidization of char using a gasifying agent. This results in the production of hydrogen-enriched syngas in biomass gasification. Using steam as a gasifying agent increases both the methane content and hydrogen concentration in syngas [31].

3.3. Entrained Flow (EF) Gasifiers

In an EF gasifier, gas and biomass particles move simultaneously, with low residence time (Figure 2(iii)). EF gasifiers can also handle atomized liquid feedstocks or slurry. Due to the elevated operating temperatures, all EF gasifiers generate slag. Oxygen is mainly utilized as the oxidizing agent in EF gasifiers. These gasifiers operate at higher temperatures (1200–1600 °C), resulting in minimal tar production due to high thermal cracking and low CH₄, maximizing the production of pure syngas and producing significantly little biochar. Moreover, hydrogen and CO₂ are produced more readily at higher temperatures compared to methane. EF gasifiers can process a diverse range of feedstocks as the particle size is suitable and the composition remains consistent over time [32].

3.4. Plasma Gasifiers

In plasma gasifiers, high-voltage discharge between graphite electrodes generates the plasma and acts as a heat source. Temperatures ranging from 6000 to 10,000 °C enable the arc to convert hydrocarbon gases, liquids, and solids into a gaseous mixture of H₂ and CO (Figure 2(iv)). While consistent gasifier conditions are maintained, the syngas concentration is controlled by adjusting the plasma current. All organic materials are converted into high-quality syngas, while the remaining inorganic materials are vitrified into inert slag [33,34]. A plasma gasifier yields very little or no biochar due to its high temperature, which promotes complete biomass conversion.

3.5. Hydrothermal Gasifiers

A hydrothermal gasifier operates at 550 °C and 40 bar, producing syngas rich in hydrocarbons and/or hydrogen. This method converts wet biomass into syngas. It utilizes water from the biomass as a reaction medium, and during the supercritical phase, it produces syngas (Figure 2(v)). Like supercritical water oxidation, which operates in the supercritical region but requires higher temperatures due to exothermic oxidation reactions, this process employs temperatures that exceed the critical point of water [35,36]. Tar is an inevitable product in any kind of biomass gasifier. However, the syngas from downdraft gasifiers and updraft gasifiers typically contains tar contents of 1 g Nm⁻³ and up to 100 g Nm⁻³, respectively [37]. The presence of tar in syngas prevents its efficient usage in engines. Tar content may be minimized by chemical and physical methods, viz., a catalytic or downstream processing approach. The chemical method is preferred to reduce tar content and to improve syngas yield and fuel quality [38,39]. It produces minimal biochar due to a high-pressure, supercritical environment for gasification.

The type of gasifier used significantly impacts the performance of biomass gasification and the production of tar due to distinctions in design, temperature profiles, and residence

times. In updraft gasifiers, producer gas moves through low-temperature pyrolysis and drying sections before being released, which results in a higher tar content in the producer gas. In contrast, downdraft gasifiers, which feature a co-current flow of air and fuel, enable tar to travel through high-temperature oxidation and reduction zones, reducing tar content in the producer gas. Moreover, fluidized bed gasifiers (particularly circulating bed gasifiers) provide consistent temperature distribution and rapid heat exchange, facilitating tar cracking and producing syngas with minimal tar [40,41]. Hence, selecting a suitable type of gasifier is essential for enhancing gasification performance and reducing tar formation.

4. Tar Cracking Mechanisms in Catalytic Biomass Gasification

Tar is a mixture of oxygenated organic components with a molecular weight greater than that of benzene (boiling point $> 150\text{ }^{\circ}\text{C}$). Tar is produced when raw biomass materials undergo partial reaction and condense on metal surfaces at ambient temperatures. Tar contains primary elements (C and H) and secondary elements (O, N, and S). Biomass tar is a complex and diverse substance, comprising approximately 10,000 different organic compounds, including sulfur-containing hydrocarbons, aromatic compounds, oxygen-containing hydrocarbons, and polycyclic aromatic hydrocarbons (PAHs). It poses a carcinogenic hazard to the environment [42]. Varieties of catalysts are used in biomass gasification to minimize tars in syngas.

For gas-phase thermal cracking reactions, the composition of tars generated by biomass gasification can be divided into four primary categories: Products made from cellulose-based materials, such as furfurals, hydroxy acetaldehyde, and levoglucosan, are examples of primary tars; olefins and phenols are secondary tars; methyl derivatives of aromatic compounds like indene, methylnaphthalene, methylace naphthylene, and toluene are tertiary tars; benzene, naphthalene, acenaphthylene, anthracene/phenanthrene, and pyrene are a few of the condensed tertiary products. Tar is formed at different gasifier reaction temperatures (Figure 3). Biomass is gasified independently for ex situ catalytic biomass gasification, and the gas produced is moved to the catalyst bed after gasification [43]. The catalyst and biomass to be gasified are combined in in situ catalytic biomass gasification [25].

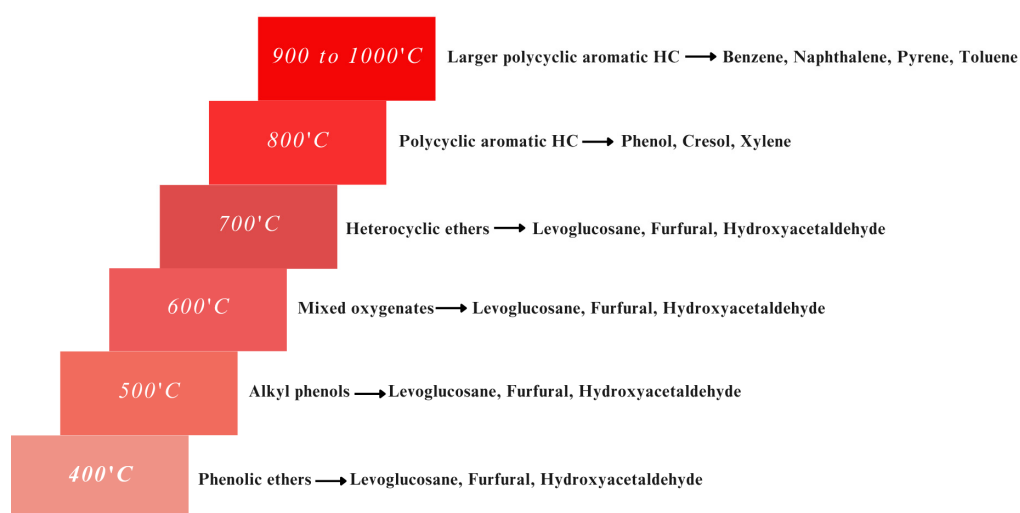
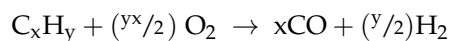
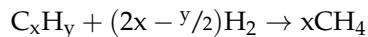
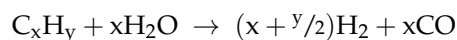
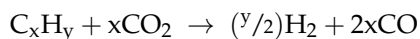


Figure 3. Tar compounds at varied temperatures in a gasifier.

Tar cracking is the process that converts higher, denser, and more complex organic compounds of tar into simpler, light molecules without using hydrogen. It is accomplished by using heat and a catalyst [44]. The following chemical reactions occur throughout the

processes of dry reforming, steam reforming, hydrogenation, thermal cracking, carbon production, and partial oxidation that lead to tar degradation:



Along with the breakdown of dealkylated side chains, molecular cyclization of hydrocarbons, and aromatization events, the tar transformation process entails the destruction and transformation of heteroatom-containing active groups in the tar [45]. In gasification, heteroatoms (oxygen, nitrogen, and sulfur) in the biomass form stable aromatic and heterocyclic organic compounds during the pyrolysis stage. These molecules have a larger heteroatom content than pure hydrocarbons because they condense into tar and resist total decomposition. Catalytic gasification of biomass significantly reduces the concentration of tar through reforming. As the process is endothermic, some of the chemically bound energy in the gas must be oxidized to sustain the process. Thus, the gasification process becomes less efficient. Methane, ethane, and propane are among the low hydrocarbons left intact when the catalyst reforms the tar content [46].

Effective catalytic processes for tar cracking in biomass gasification play a role in enhancing syngas purity and operational efficiency by mitigating issues such as equipment fouling and reduced gasification performance. Understanding the complexities of tar composition and employing appropriate catalytic strategies are essential for optimizing catalytic biomass gasification technologies.

5. Catalytic Biomass Gasification

The composition of tar in syngas primarily depends on biomass properties and the gasification environment. The conditions might be particle size, particle size distribution, elemental composition, minerals, the moisture content of the feedstock, the type of gasifier, and the gasifying agent used [47]. In general gasification, during the tar cracking process, tar is converted into fuels at higher temperatures (representing an energy-intensive process) due to the requirement of higher activation energy. This higher energy requirement for tar cracking can be reduced by adding a catalyst [48].

A catalyst is a substance that accelerates a chemical reaction when introduced in small amounts, without undergoing any permanent chemical changes itself. Efficient catalysts shorten reaction time and temperature, reduce the activation energy needed for gasification processes, and achieve high carbon conversion rates. Catalyst addition not only reduces reaction temperature but also results in good-quality syngas. When appropriate catalysts are used, the gasification rate is increased, and the reaction temperature is lowered. Notably, the tar formation and tar quantity decrease with an increase in the reaction temperature of the gasification process. It is also noted that tar quantity varies with the gasifying agent. The tar content was lower for air gasification than for steam gasification [49]. Catalysts can also be employed to either encourage or inhibit the synthesis of gaseous product constituents. A meager quantity of methane, which is a mixture of H_2 and CO , is also formed during syngas production. Catalytic gasification can either stimulate or inhibit methane generation [50]. For example, nickel-based catalysts stimulate methane production through methanation

reactions, especially under moderate temperatures and hydrogen-rich conditions, and an alkali metal catalyst (K_2CO_3) inhibits methane formation by promoting tar cracking and syngas production (CO , H_2) over methanation. It is observed that the calorific value of syngas is reduced if it contains tar [49].

Alkali metal salts of weak acids, such as potassium carbonate (K_2CO_3), sodium carbonate (Na_2CO_3), potassium sulfide (K_2S), and sodium sulfide (Na_2S), can be used in steam biomass gasification. Catalytic gasification has drawbacks, such as higher catalyst material costs (mostly related to precious metals), a gradual decline in catalyst efficacy, and issues in regeneration and recycling [51]. Catalytic regeneration can be both advantageous and disadvantageous. While it extends the catalyst lifespan and reduces waste, it can be costly and energy-intensive. It may not fully restore the original activity (structural degradation during regeneration), especially for catalysts like ruthenium that degrade quickly without stabilizing supports. Potassium carbonate is a versatile catalyst and may be recovered from used coke by washing it with water [52]. Poisoning, alongside aging, is another factor that can degrade catalysts. Many catalysts are vulnerable to specific substances that adhere to them or alter their composition to make them ineffective [53]. When a catalyst is present, reactions such as tar cracking, steam reforming, thermal cracking, dry reforming, and steam dealkylation occur. The process of tar degradation releases gases such as H_2 , CO , CH_4 , and CO_2 [42].

There are two approaches to modifying the catalytic conversion of tar: To initiate in situ catalytic gasification, which involves catalyzing tar conversion within the reactor, the first method is to mix the primary catalyst with the biomass feedstock. Alternatively, the second method reduces the tar content in the syngas by employing a separate catalytic reactor (secondary catalyst) located downstream of the gasifier. Although the latter strategy offers better tar removal efficiency, it can be costly and complex for small and medium-sized gasification systems. In contrast, in situ tar cracking provides a more feasible and economical approach [53].

Catalysts may be added to feedstock through wet impregnation or dry mixing. The results of this approach revealed that catalyst recovery became more complex and thus increased syngas production costs. Suitable catalysts are (i) efficient in tar cracking/removal, (ii) resistant to carbon fouling and sintering-induced deactivation, (iii) easily regenerable, and (iv) inexpensive [54]. The catalytic gasification of algal cake generated the highest total syngas yield (80.21 wt%) with the maximum H_2 yield of 8.22 mmol g^{-1} biomass using a Ni catalyst [11]. The gasification of dry food waste produced a total gas yield of 7.89 mol.kg^{-1} . The H_2 yield from the non-catalytic and catalytic ($NaOH$) gasification of food wastes was 2.0 mol.kg^{-1} and $12.73 \text{ mol.kg}^{-1}$, respectively [55].

In general, catalysts play a critical role in enhancing the output of gasification processes by accelerating reaction rates, lowering reaction temperatures, and minimizing tar production. Despite their effectiveness in influencing syngas composition and improving efficiency, challenges such as high material costs, a gradual loss of activity, and complexity in regeneration and recycling emphasize the ongoing need for advancements in catalyst technology.

6. Types of Catalysts Used in Biomass Gasification

Various catalysts are used to reduce the tar content in syngas and lower the reaction temperature for gasification [56]. The catalysts used in the gasification process (Figure 4) are categorized into six groups [57].



Figure 4. Catalyst groups.

6.1. Alkali and Alkaline Earth Catalysts

Alkali and alkaline earth metals that occur naturally in biochar and functional groups containing O₂ cause tar to reduce or decompose significantly. Depending on the biomass type, the inorganic fractions of raw biomass or charcoal can include different amounts of Si, Al, P, K, and Ca. Significantly, biomass has larger quantities of K, Ca, and Mg, inorganic components that influence the reactivity of gasification. Due to the presence of trace amounts of potassium aluminosilicate, magnesium silicate, sodium aluminosilicate, sodium calcium aluminosilicate, and SiO₂, magnesium-rich granular bed materials help to prevent agglomeration. Therefore, this effect can be used to enhance the reaction process. For example, biomass feedstocks with a rich alkali metal content can perform better than granular beds [58]. Alkali and alkaline earth metals are well-suited for gasifying energy crops, agricultural residues, and forest residues. They are effective in reducing tar formation during gasification and result in a high hydrogen content in the syngas. They can be used in large-scale, cost-effective applications.

6.2. Transition Metal Catalysts

A transition metal forms one or more stable ions while partially filling its d-orbitals. Transition metals can exchange electrons with other compounds and change their oxidation states by leaving their d orbitals empty. As a result of active element states, tar degradation proceeds more efficiently. In the gasification process, iron, cobalt, copper, nickel, and their derivatives are frequently utilized as catalysts. Because transition metal catalysts have characteristics that support both heterogeneous and homogeneous catalysts, they are suitable to be used as hybrid catalysts. Higher energy and surface area make transition metal nanoparticles active catalysts [59].

The total surface area and activity of these catalysts are negatively impacted by the coking and sintering of relatively large metal particles. New metal catalysts, such as Rh, Ru, Pd, Pt, etc., have been employed in the biomass gasification process to lessen tar content. These catalysts work well to separate tar from syngas, but they cost a lot more than traditional catalysts and nickel. Various transition metal catalysts are utilized in the gasification process, such as nanoscale Zn-doped MgO catalysts, olivine ((Mg, Fe)₂SiO₄), and composite catalysts (KCo, KNi, KFe, and KCe) [60,61]. They can be used in the gasification of agricultural residues and wood processing residues. They can be reused multiple times, so they enhance economic feasibility, along with tar reduction and high-quality syngas production.

6.3. Ni-Based Catalysts

With unique qualities such as higher activity, reduced cost, and the ability to promote hydrogen reactions and water–gas transfer, nickel-based catalysts are the most suitable catalysts for tar reforming and cracking in syngas. Furthermore, they improve the quality of the syngas generated during the gasification process. However, coking and sulfur poisoning deactivate nickel active sites. Ni nanoparticles can be dispersed on an alkali-metal-doped support or doped with carbon to address these issues. In gasification, metal catalysts such as Ni/CeO₂/Al₂O₃, Ni/MgO, Ni/Al₂O₃, Ni/ZnO, NiO/γ-Al₂O₃, and Ni/ZrO₂ are utilized [59,62]. These catalysts can be effective in the gasification of agricultural residues and energy crops. Their advantages may be a reduced tar yield and high hydrogen content in the syngas with enhanced thermal and chemical stability.

6.4. Carbon-Based Catalysts

Biochar possesses several remarkable qualities, including a large specific surface area, porous structure, functional groups, high reliability, affordability, and ease of deactivation and recovery. The biochar produced from the gasification process as a byproduct and the pyrolysis process as a main product has significant potential for numerous applications [63]. Biochar can act as a supporting medium to immobilize nanocatalysts to increase their stability and performance. It can offer high tar removal efficiency and a sustainable option in syngas production from the catalytic biomass gasification of agro-residues, algae, and wet waste.

6.5. Natural Mineral Catalysts

Dolomites are naturally occurring minerals made up of magnesium and calcium carbonates that can break down into oxides at higher temperatures. Trace minerals, including SiO₂, Fe₂O₃, and Al₂O₃, may be present in dolomites and other naturally occurring catalysts; iron oxide is crucial for catalytic activity [64]. As natural mineral catalyst precursors, borax and dolomite are employed. The natural mineral catalyst olivine is made up of silica, iron oxide, and magnesium oxide [65]. It is compatible with the gasification of agricultural residues and forest residues. It is cost-effective but results in moderate tar removal.

6.6. Catalysts Derived from Waste Byproducts

While catalysts made of nickel and noble metals are very effective at cracking tar, they can also be costly and create hazardous byproducts. Natural materials are an alternative that addresses these issues as they are inexpensive catalysts [66]. Waste materials originating from organic or inorganic sources can be generated through different processes and can be used as catalysts in gasification. Therefore, the use of catalysts derived from waste substances is a viable option for minimizing waste and lowering environmental implications due to the disposal problem. These catalysts may be rice husks, red mud, aluminum slag, fly ash, iron slag, sludge, snail shells, eggshells, seashells, coconut shells, and waste from gold mines [67]. They are suitable for gasification of agricultural residues and wet waste. They offer a sustainable and environment-friendly option to reduce the operational cost and enhance the tar reduction in syngas.

In conclusion, catalysts play a pivotal role in biomass gasification by enhancing reaction rates, lowering activation energies, and facilitating tar reduction. Various types of catalysts, including alkali metals, transition metals, nickel-based catalysts, carbon-based materials, natural minerals, and waste byproducts, offer diverse approaches to improving syngas quality and overall process efficiency. Each catalyst type presents unique advantages and challenges, influencing its suitability based on specific operational requirements and economic considerations in biomass gasification systems.

7. Yield and Composition of Syngas from Catalytic Biomass Gasification

The gas yield, syngas composition, and tar content of biomass gasification may be affected with or without a catalyst. Generally, non-catalytic biomass gasification syngas has a higher CH_4 content than catalytic biomass gasification syngas [68]. Catalytic biomass gasification offers a higher syngas yield, and the addition of a catalyst enhances the catalytic activity, resulting in higher syngas purity than non-catalytic biomass gasification. The catalyst type and catalyst-to-biomass ratio (CBR) used in the process have a significant impact on the syngas yield. Numerous studies have been carried out using different catalysts to produce high-purity tar-free syngas. Table 1 shows the syngas yield and composition from catalytic biomass gasification.

Syngas synthesis in a dual fluidized bed gasifier using a silica–alumina catalyst can eliminate tar and produce high-quality syngas from biomass. The gasification process was conducted at 500 to 750 °C using steam as an oxidizing agent. The SBR (steam-to-biomass ratio) and the CBR (catalyst-to-biomass ratio) were 1 and 10–20, respectively. The solid–solid reaction process enhanced the H_2 -to- CO molar ratio by more than 10 [69].

A downdraft fixed bed gasifier and a fluidized bed gasifier were used to generate hydrogen-rich syngas from pine sawdust with an air–steam mixture and dolomite and nickel-based catalysts. Under the optimized conditions, the maximum recovery of H_2 was 52.47% (vol), with a H_2/CO ratio of 1.87–4.45 [56]. The two-stage catalytic pyrolysis and gasification of pine sawdust was performed using several nickel-based catalysts to produce energy-rich syngas. A high recovery of $3.29 \text{ N m}^3 \text{ kg}^{-1}$ biomass syngas was generated at 850 °C [69]. Three different kinds of catalysts, i.e., limestone, olivine, and calcined dolomite, were utilized to increase syngas yield and tar reduction. In a fluidized bed gasifier, biomass was gasified using a catalytic air–steam process. Using calcined dolomite, the highest H_2 mole fraction of 49.1% (vol) was achieved at 1000 °C [70]. The potential of obtaining high-purity syngas from the gasification of birchwood with calcined dolomite was examined in a fluidized bed gasification system [71]. A high H_2/CO ratio was obtained at 700–800 °C.

Table 1. Syngas yield and composition from catalytic biomass gasification.

Biomass	Reactor	Oxidation Agent	Temp, °C	Catalyst	Catalyst Loading	Yield	HV, MJ Nm ^{−3}	Syngas Composition, %				Other Details	Ref.
								H ₂	CH ₄	CO	CO ₂		
Banana	Fixed bed	Steam	368	Ni/Al ₂ O ₃	2.5% w/w	Syngas: 178 mg g ^{−1}	15.21 (LHV)	51.8	0.44	0	22.54	-	[72]
Microalgae (<i>C. vulgaris</i>)	Horizontal tubular reactor	Air 20 mL min ^{−1}	851	ZnOeNieCaO	16.4 wt%	Gas: 83.34 wt%; Char: 10.34 wt%; Tar: 6.32 wt%	14.86 (LHV)	48.95	10.12	18.27	22.64	Residence Time: 28.8min	[73]
Pine sawdust	Fluidized bed	SBR: 1.0	650	Fe-based composite	CBR: 1.2	Gas: 60.4% Char: 37.1% Tar: 2.5%	-	42.2	16.5	22.5	16.4	-	[65]
Citrus peel residues	Fixed bed	SBR: 1.5 wt%/wt%	750	Dolomite	-	-	-	63	1	16	20	CCG: 54.9% HCE:54.1%	[74]
Pine sawdust	Fluidized bed	Air-steam Air: 0.65 Nm ³ h ^{−1} Steam: 0.4 ER: 0.3, 0.2 SBR: 0.85, 0.75	800	Dolomite	65 g	Gas: 1.54 Nm ³ kg ^{−1} ; Tar: 19.05 g kg ^{−1}	-	38.38	7.02	24.89	27.62	-	[56]
					56 g	Gas: 1.56 Nm ³ kg ^{−1} ; Tar: 13.85 g kg ^{−1}	-	38.13	7.48	26.06	26.20		
Pine sawdust	Downstream fixed bed	Air-steam Air: 0.65 N m ³ h ^{−1} Steam: 0.4 ER: 0.3 SBR: 0.85	800	Dolomite	56 g	Gas: 2.24 Nm ³ kg ^{−1} biomass; Tar: 4.29 g kg ^{−1} biomass	-	50.23	4.3	12.3	32.56	Residence Time: 28.8 min	
Wood residue	Fluidized bed	ER:0.17 SBR: 0.71 wt%/wt%	823	Ni/Al ₂ O ₃	40%	Gas: 90.33%; Tar: 4.72%; Char: 04.95%	-	36.17	11.12	24.26	24.25	Residence Time: 26 min; CCE: 86.17%; CGE: 56.24%	[19]
Wood residue	Fluidized bed	ER: 0.17 SBR: 0.71 wt%/wt%	823	Ni/CeO ₂ /Al ₂ O ₃	40%	Gas: 96.84%; Tar: 2.94%; Char: 0.78%	-	42.52	11.47	23.04	18.10	Residence Time: 44 min; CCE: 93.65%; CGE: 71.6%	
Wood chips/coconut shell	Downdraft gasifier	Air flow rate: 400 L min ^{−1} BR: 70:30	-	Dolomite	10%	-	4.96 (HHV)	10	2	22.5	10	-	[75]

Table 1. Cont.

Biomass	Reactor	Oxidation Agent	Temp, °C	Catalyst	Catalyst Loading	Yield	HV, MJ Nm ^{−3}	Syngas Composition, %				Other Details	Ref.
								H ₂	CH ₄	CO	CO ₂		
Enteromorphain testinalis	Fluidized bed	ER: 0.14 SBR: 0.5–1.0 wt%	800–1000	Dolomite	-	Gas: 90.2%; Tar: 4.3%; Char: 5.5%	11.6 (HHV)	49	1	27	23	Residence Time: 50 min; CCE: 60.8%; GE: 71.5%	[70]
Enteromorphain testinalis	Fluidized bed	ER: 0.14 SBR: 0.5–1.0 wt%	800–1000	Olivine	-	Gas: 88.5%; Tar: 5.5%; Char: 6.0%	12.5 (HHV)	48	1	25	26	Residence Time: 50 min; CCE: 58%; GE: 69.1%;	
Enteromorphain testinalis	Fluidized bed	ER:0.14 SBR: 0.5–1.0 wt%	800–1000	Lime	-	Gas: 85.5%; Tar: 6.1%; Char: 8.4%	14.2 (HHV)	49	1	30	20	Residence Time: 50 min; CCE: 51.8%; GE: 60.5%	
Pine sawdust	Tube furnace reactor	-	850	Nickel-based catalysts	-	Gas: 2.78 Nm ³ kg ^{−1} biomass	9.6 (LHV) 10.9 (HHV)	2.25 Nm ³ kg ^{−1}	0.54 Nm ³ kg ^{−1}	-	-	-	[69]
Wood chips of red pine	Two-stage fluidized bed reactor	SBR: 3 mol/mol	600° C	Ni/Al ₂ O ₃	-	Gas: 62 mmol/g-daf	14 (LHV)	36.2	2.6	10.3	14	-	[76]
Wood chips of red pine	Two-stage fluidized bed reactor	SBR: 3 mol/mol	600	Ni/BCC	-	Gas: 2 Nm ³ kg ^{−1} ; Tar: 60 mg Nm ^{−3}	14 (LHV)	46.6	4.3	21.6	16	-	
Pig manure compost	Two-stage fluidized bed reactor	SBR: 3 mol/mol	600	Ni/BCC	-	Gas: 54 mmol/g-daf	14 (LHV)	26	5	12	11	-	
Rice husk	Bubbling fluidized bed	ER: 0.10	850	Calcined dolomite	-	Gas: 80.4 wt%; Char: 11.2 wt%; Tar: 7.5 wt%	12.2 (LHV)	34.1	4.9	35.0	22.3	-	[77]

A study was conducted in a fluidized bed gasifier with wood residue as a feedstock and Ni/CeO₂/Al₂O₃ and Ni/Al₂O₃ as metal catalysts for maximizing syngas and hydrogen production. The process operational parameters considered were 750 to 900 °C with a 20–60 min residence time and 20–40% catalyst loadings. The maximum syngas and hydrogen yield was attained with a 60 min residence time. For Ni/Al₂O₃ and Ni/CeO₂/Al₂O₃ catalysts, the maximum mole fraction of syngas (H₂ + CO) was 44.93–60.43% and 58.17–65.56%, respectively. The maximum tar reduction compared to the non-catalytic process was 162% and 196% when Ni/Al₂O₃ and Ni/CeO₂/Al₂O₃ were used as catalysts [19].

The syngas yield was 178 mg g^{−1} and the hydrogen content was 51.8% (mol) when 2.5% *w/w* of a Ni/Al₂O₃ catalyst was used with banana waste as a feedstock in a fixed bed gasification system. Interestingly, the CO₂ production was halved, and the CO content declined from 3.87 to 0% (mol) [72]. Catalytic co-gasification of wood chips with two coconut wastes was studied in a downdraft gasifier resulting in a high syngas yield. For this purpose, two catalysts, i.e., limestone and dolomite, were used to study their influence on syngas composition. The study resulted in dolomite (30% loading) yielding the maximum syngas yield with H₂ (10%), CO (22.5%), CH₄ (2%), and CO₂ (10%) [75].

Another study evidenced that the use of a natural eggshell-based CaO catalyst improved the catalytic activity of the catalytic gasification of biomass. It suppresses CO₂ generation by its absorption ability, which stimulates H₂ production over a low reaction temperature by causing the water–gas shift reaction. The process was carried out at 500–1000 °C, with a flow rate of 10 mL min^{−1}, in a 5% O₂/He environment. Wood gasification was conducted at varied CaO catalyst loadings (20, 40, and 60%), and the yield of H₂ was increased by 57, 60, and 73%, respectively, than non-catalytic gasification process [78]. Furthermore, two lipid-extracted microalgae biomasses were gasified using a thermogravimetric analyzer at temperatures ranging from 30 to 800 °C in a 5% O₂/Ar atmosphere, at a flow rate of 500 mL min^{−1}, with a waste eggshell-derived CaO catalyst. Increased catalyst loading (10, 30, and 50%) increased the H₂ concentration from 21.2 to 114.4% and decreased the CO and CO₂ contents by 54 and 53%, respectively, compared to the non-catalytic gasification of microalgae [79].

Catalysts significantly enhance biomass gasification processes by improving syngas yield, composition, and purity and reducing tar content. Studies exploring various catalyst types and operational parameters underscore catalysts' potential to optimize syngas production efficiency, thereby advancing the feasibility of gasifying biomass as a sustainable energy source.

8. Current Status and Future Perspectives

Biomass's affordability, abundance, and carbon-neutral nature encourage the usage of catalysts in various thermochemical processes to synthesize high-value products. Consequently, there is a significant need to develop innovative and efficient catalysts. Future advancements are expected to enhance efficiency and eco-friendly benefits and explore novel applications. Catalysts reduce energy demands for gasification while increasing syngas production from biomass [80]. Several parameters play a significant role in gasification and can be used to optimize the process conditions and obtain higher yields of energy/fuel from biomass feedstocks. Studies are focusing on the improvement of catalysts. The key focus areas and potential challenges related to catalytic biomass gasification are illustrated in Figure 5.



Figure 5. Researchable issues in the catalytic biomass gasification process.

The critical parameters for the catalysts used in the gasification process are sintering, sulfur poisoning, coking resistance, and low cost. Catalyst deactivation requires in-depth analysis for a better understanding of its role in tar cracking mechanisms. Sometimes, the catalysts can be deactivated due to various factors such as poisoning, contamination, thermal degradation, the formation of vapor-phase compounds, vapor–solid or solid–solid interactions, abrasion, and attrition [81]. Used catalysts may be disposed of, but their lifespan can be extended if they are recycled. While catalytic disposal may seem economically advantageous, its environmental impact due to toxic composition must be considered. Therefore, sustainable approaches for catalyst regeneration that are practical, reliable, and cost-effective need to be developed. In catalytic biomass gasification, challenges include developing stable catalysts with enhanced selectivity and integrating catalysis with separation processes [82]. Despite advancements, toxicity and catalyst recovery from mixtures continue to limit catalysts' widespread applications. Finally, the lifecycle assessment of catalytic biomass gasification may be thoroughly studied to assess its impact on the environment.

9. Conclusions

Recently, green hydrogen generation using the biomass gasification route has become more attractive for fulfilling future energy needs. The gasification process can generate syngas from a wide range of biomass feedstocks. This gasification is not commercialized due to corrosive tar in syngas and its associated problems in applications. Catalytic gasification provides tar reduction and syngas quality enhancement, enabling syngas utilization for wider applications. Catalysts commonly used in biomass gasification for this purpose include waste byproducts, natural mineral catalysts, alkali and alkaline earth metal catalysts, and nickel-based catalysts. To optimize the syngas synthesis process and yield, it is essential to investigate the advantages and disadvantages of each of these catalysts. The catalysts that are currently in use need to become more stable, efficient, inexpensive, sustainable, and reusable and contribute to less energy-intensive processes to create a more environmentally friendly and sustainable future. Ongoing research on catalyst design aims to resist deactivation caused by poisoning, sintering, and carbon deposition in the gasification process. Also, effective regeneration techniques are needed to catalyze restoration without a loss of efficiency or material. Developments in this research area will lead to reduced operational costs, process stability enhancement, and continuous long-term catalyst utilization, eventually making catalytic biomass gasification more feasible and

sustainable for wider industrial applications. Therefore, it is necessary to synthesize a suitable catalyst from organic substances, especially for biomass gasification, as this may be beneficial for the dissemination of this technology at a commercial scale.

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Abbreviations

The following abbreviations are used in this manuscript:

BFB	Bubbling Fluidized Bed
CFB	Circulating Fluidized Bed
DFB	Dual Fluidized Bed
EF	Entrained Flow
PAHs	Polycyclic Aromatic Hydrocarbons
CBR	Catalyst-to-Biomass Ratio
SBR	Steam-to-Biomass Ratio
CBR	Catalyst-to-Biomass Ratio
HV	Heating Value
LHV	Lower Heating Value
HHV	Higher Heating Value
CCE	Carbon Conversion Efficiency
GE	Gasification Efficiency
CGE	Cold Gas Efficiency
HCE	Hydrogen Conversion Efficiency

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